

Gregory Bian

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EDUCATION

University of British Columbia

Bachelor of Applied Science, Computer Engineering (Co-op)

Expected May 2029

Vancouver, BC

TECHNICAL SKILLS

Programming & Frameworks: C, C++, Python, Java, JavaScript

Systems & Tools: Git, Docker, Linux (Ubuntu), Pytest, JUnit

Frameworks and Libraries: React, Flask, NumPy, Pandas, Matplotlib, OpenCV, BeautifulSoup/Selenium

TECHNICAL EXPERIENCE

Firmware Developer

UBC Solar

Sep 2025 – Present

Vancouver, BC

- Collaborating with a cross-disciplinary team of **80+** software, electrical, and mechanical engineers, writing firmware for a solar-powered car, Cascadia, in preparation for FSGP and ASC solar racing competitions.
- Developing firmware for a **STM32**-based Driver Dashboard in **C**, implementing multi-page functionality on the LCD Screen to improve visibility of real-time driving and diagnostic data.
- Implementing memory-robust, **FreeRTOS**-based task scheduling to prevent race conditions and ensure predictable and reliable behavior to prevent firmware faults at competition.
- Integrating **CAN** bus communication between ECU's, enabling quick and reliable real-time telemetry transfer.

Software Developer

UVIC Aero

Jan 2025 – Jun 2025

Victoria, BC

- Configured a **Linux**-based flight simulator to test and validate autonomous flight functions for a fire-extinguishing drone, Firefly, ensuring seamless integration with the mechanical subsystem.
- Engineered a **Python**-based hotspot detection algorithm for infrared images using thresholding and connected components analysis in **OpenCV**, identifying over **85%** of high-intensity regions in test datasets of **30+** images.

STEM and Computing/Robotics Instructor

UBC Geering Up Outreach

Jun 2025 – Aug 2025

Vancouver, BC

- Led STEM-based Summer Camps for Geering Up, a **STEM Outreach Program** at the Faculty of Applied Science at UBC, reaching over **28000+** youth a year.
- Instructed up to 30 students per week on **Arduino**-based embedded programming, computer architecture, and data analysis in **Python**, explaining complex technical concepts in a fun and engaging way for youth.

PROJECTS

NotiFlow Course Tracker | *JavaScript (React), Flask, Python*

Nov 2025 – Dec 2025

- Led a team of 5 in developing a Course Tracking Chrome Extension; implemented a **React**-based UI and integrated Chrome Local Storage APIs to store data locally, improving usability and reliability.
- Leveraged **Flask** routing to send and receive data between the Chrome Extension and backend Python scripts and support features such as Canvas API access, BeautifulSoup webscraping and OpenAI API prompt filtering.

OpenCV Mini Sudoku Solver | *Python, OpenCV, Numpy*

Aug 2025 – Sep 2025

- Developed a **Python** script using **OpenCV** to automatically solve Mini Sudoku puzzles by implementing image thresholding, grid detection, digit recognition, and algorithmic solving, resulting in a **95%** accurate solution.
- Resolved Optical Character Recognition issues (ie. digit misclassification, noise in blanks) through iterative experimentation with contour detection, thresholding, and adaptive cropping.

Hand Hygiene Compliance System | *MicroPython, JavaScript, Flask, Raspberry Pi*

Mar 2024 – Apr 2024

- Engineered a prototype embedded system with **Raspberry Pi Pico** in **MicroPython** to monitor Hospital Hand Hygiene Compliance of workers, resulting in a **30%** higher efficiency compared to alternate designs.
- Developed a full-stack user interface in **HTML/CSS/JavaScript**, featuring real-time compliance data visualization and an interactive reset button with two-way communication with **Flask** to backend systems.